

Historical paintings recreated using deep learning and 3D printing

A team of scientists from Massachusetts Institute of Technology's (MIT's) Computer Science and Artificial Intelligence Laboratory (CSAIL) has created a system aimed at designing reproductions of world-famous paintings.

The new system—called RePaint—uses a combination of 3D printing and deep learning to authentically recreate historical paintings, regardless of different lighting conditions or placement. RePaint can be used to remake artwork for

a home, protect originals from wear and tear in museums, or even help companies create prints and postcards of such masterpieces.

To test RePaint, the team reproduced a number of oil paintings created by an artist collaborator. It found that the system was more than four times more accurate than state-of-the-art physical models at creating exact colour shades for different artworks.

As of now, reproductions are only about the size of a business card, due to the time-intensive nature of printing. In the future, the team expects that more advanced, commercial 3D printers could help with making larger paintings more efficiently.



RePaint system reproduces paintings by combining two approaches called colour-contoning and half-toning, as well as a deep learning model focussed on determining how to stack 10 different inks to recreate specific shades of colour (Credit: MIT's CSAIL researchers)

System that provides solar- or wind-generated power on demand

Engineers from MIT have come up with a conceptual design for a system to store renewable energy, such as solar and wind power, and deliver that energy back into an electric grid on demand. The system may be designed to power a small city not just when the sun is up or the wind is high but around the clock.

The new design stores heat generated by excess electricity from solar or wind power in large tanks of white-hot molten silicon, and then converts the light from the glowing metal back into electricity when needed.

Researchers estimate that such a system would be vastly more affordable than lithium-ion batteries, which have been proposed as a viable, though expensive, method to store renewable energy.

They also estimate that the system would cost about half as much as pumped hydroelectric storage, which is the cheapest form of grid-scale energy storage till date.



Concentrated solar plants store solar heat in large tanks filled with molten silicon, which is heated to high temperatures of about 537.8°C (1000°F). When electricity is needed, the hot silicon is pumped through a heat exchanger, which transfers the silicon's heat into steam. A turbine then turns that steam into electricity (Credit: MIT)